12-12-2019(E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Max. Marks: 100 | Nov-Dec 2019 **End Semester Exam** | **Duration:3 hours**

Class: S.Y. Btech | **Semester: IV**

Name of the Course: Analysis of Algorithm | **Branch: COMP**

Course Code: UCEC403

Instructions:

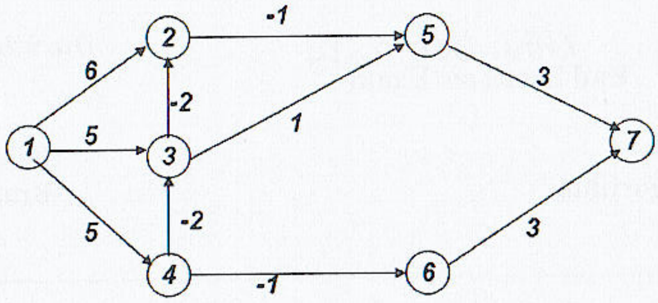
(1) All Questions are Compulsory.

(2) Figures to the right indicate full marks.

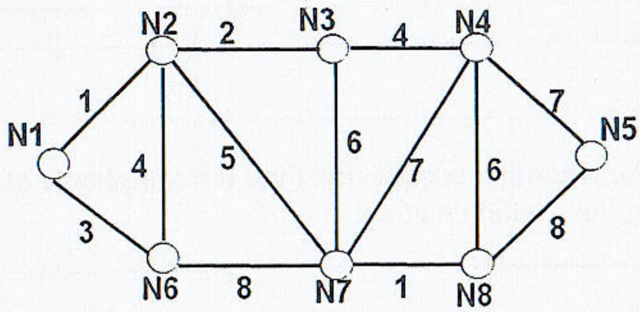
(3) Assume suitable data, if necessary.

<table>
<tr><th>Question No.</th><th></th><th>Max. Marks</th></tr>
<tr><td>Q1(a)</td><td>Define θ, O, Ω notations for algorithm complexity. Find the complexity of following recurrence using Substitution method
$T(n)=2T(n-1)+1$</td><td>10</td></tr>
<tr><td>Q1(b)</td><td>Explain Master's Method for solving recurrences.Solve the given recurrences using Master Theorem-
a. $T(n) = 16T(n/4) + n^2$
b. $T(n) = T(2n/3) + 1$</td><td>10</td></tr>
<tr><td>Q.2(a)</td><td>Explain Dynamic programming approach to solve knapsack problem. Solve the following Knapsack Problem
Let n=4, W_i={3, 4, 5, 6}, P_i={50, 40, 10, 30}, Knapsack size M=10
OR
Find the optimal solution for matrix chain multiplication with dimension sequence : <15, 8, 10, 12, 3></td><td>10</td></tr>
<tr><td>Q.2(b)</td><td>Write Quick Sort Algorithm and derive the best case and worst case complexity of it.
OR
Write an algorithm for Merge Sort. Derive the complexity of Merge sort.</td><td>10</td></tr>
<tr><td>Q.3(a)</td><td>Find all pairs shortest path for the given weight matrix using Floyd-Warshall's Algorithm
<table>
<tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr>
<tr><td>A</td><td>0</td><td>5</td><td>8</td><td>∞</td><td>-3</td></tr>
<tr><td>B</td><td>∞</td><td>0</td><td>∞</td><td>2</td><td>6</td></tr>
<tr><td>C</td><td>2</td><td>∞</td><td>0</td><td>∞</td><td>∞</td></tr>
<tr><td>D</td><td>2</td><td>∞</td><td>-4</td><td>0</td><td>∞</td></tr>
<tr><td>E</td><td>∞</td><td>∞</td><td>∞</td><td>3</td><td>0</td></tr>
</table>
OR</td><td>10</td></tr>
</table>

Consider following graph to compute shortest path from 1 to 7 using Bellman-ford Algorithm.



Q. 3(b) Apply Prim's algorithm to find minimum cost spanning tree. Mention which strategy is used to compute MCST. — **10**



Q.4 (a) Find the optimal solution for the given weight matrix using travelling sales person — **10**

	A	B	C	D
A	∞	11	6	7
B	11	∞	12	6
C	4	8	∞	10
D	11	3	1	∞

Q.4(b) Describe 0/1 Knapsack using Branch and bound with example. — **10**

OR

Write the control abstraction (General Algorithm) for

a. Greedy Algorithm Approach
b. Divide and Conquer Approach

Q.5(a) Let Text="ababababacaba" and Pattern="ababaca". Apply string matching with finite automata to above mentioned strings. Draw state transition diagram and find whether pattern is present in text or not. — **10**

Q.5(b) Find Longest Common Subsequence of X= "ABCDGH" and Y="AEDFHR". Write recursive formula for LCS. Write its complexity. — **10**